Can AI Reliably Detect AI? A Study of Fine-Tuning and Zero-Shot Performance on Academic Essays

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Abstract

The increasing use of large language models (LLMs) in educational writing contexts has led to growing interest in automated AI text detection systems. Although many AI detectors report high accuracy, it remains unclear whether these systems generalize reliably beyond benchmark datasets. This project evaluates the reliability of AI-generated essay detection using the DAIGT V2 Train Dataset alongside a small, original dataset of original human-written and AI-generated essays.

Experiments compared encoder-based classifiers and large language models under zero-shot and fine-tuned conditions, including RoBERTa-base, GPT-4.1 mini, GPT-5.4 mini, Claude Sonnet 4.6, Gemini 2.5 Flash, and Llama 3.1 8B Instruct. While several models achieved near-perfect benchmark accuracy, performance on the original dataset was not as high, suggesting that strong benchmark results did not always correlate to generalization. These findings highlight the importance of using quality data for evaluation and cautious interpretation of AI detection systems in educational settings.

1 Introduction

The rapid growth of large language models (LLMs) has transformed academic writing environments. Systems such as ChatGPT, Claude, Gemini, and Llama are now capable of generating fluent, coherent, and highly structured essays that can resemble human-written writing. As these tools become increasingly accessible, educators and institutions face growing concerns about academic integrity, authorship, and the reliability of traditional writing assessment practices.

This project investigates how well AI models classify text as human-written or AI-generated, and how reliably AI models detect AI-generated writing. I conducted experiments using the DAIGT V2 Train Dataset (TheDRCat, 2024), which contains human-written and AI-generated essays. I also created a new evaluation dataset that consists of original human-written and AI-generated essays produced specifically for this project. This project evaluates both encoder-based classifiers and LLMs under zero-shot and fine-tuned conditions, including RoBERTa (Liu et al., 2019), GPT-4.1 mini (OpenAI, 2025), GPT-5.4 mini (OpenAI, 2026), Claude Sonnet 4.6 (Anthropic, 2025), Gemini 2.5 Flash (Gemini and DeepMind, 2024), and Llama 3.1 8B Instruct (Grattafiori et al., 2024).

2 Related Work

Recent work on AI-generated text detection has shown that AI-generated writing often differs from human-written text in measurable linguistic ways. Terčon and Dobrovoljc found that AI-generated text tends to be more repetitive, more formal, and less lexically diverse than human-written writing (Terčon and Dobrovoljc, 2025). They also note that a lot of existing research focuses heavily on English GPT-generated outputs and does not always account for prompt sensitivity or cross-model variation.

Several studies have explored the use of machine learning and transformer-based models for AI-text detection. The study "Unmasking AI-Generated Texts" used linguistic and stylistic features to classify human-written and AI-generated texts and found that performance dropped on smaller custom evaluation datasets, suggesting that benchmark accuracy may not fully reflect real-world generalization (unm, 2025). The authors also observed that shorter texts are more difficult to classify because they contain fewer stylistic cues.

Research in educational assessment has similarly shown strong benchmark performance for AI-generated essay detection. Yan et al. evaluated GPT-generated essays using RoBERTa-based detectors and reported accuracies above 95% (Yan et al., 2023). However, they also emphasize the importance of validity and reliability when applying

AI-detection systems in high-stakes educational settings.

These studies motivated the focus of my project on balanced evaluation splits, source-holdout testing, and evaluation on new data. While prior work has demonstrated that AI-generated essays can often be detected successfully under benchmark conditions, my experiments investigate whether this performance generalizes reliably to more realistic educational writing settings.

3 Datasets

3.1 DAIGT V2 Train Dataset

The primary dataset used in this project was the DAIGT V2 Train Dataset from Kaggle (TheDRCat, 2024). The dataset contains 44,868 human-written and AI-generated essays from 2 human-written essay sources and 15 language models. The human-written essay sources include the PERSUADE 2.0 Corpus (Crossley et al., 2023) and the AIDE (AI Detection for Essays) dataset (Burleigh, 2023).

The DAIGT V2 Train Dataset includes a binary label for each essay, where label 0 corresponds to human-written text and label 1 corresponds to AI-generated text. Additional metadata fields include prompt name and essay source.

3.2 DAIGT V2 Split Construction

To evaluate model performance under different assumptions about dataset structure, I created three versions of the DAIGT V2 Train Dataset splits: a baseline split, a balanced split, and a source-holdout split. Each split was designed to answer a different experimental question.

3.2.1 Baseline Split

I created a baseline split as a starting point for my experiments. It uses all 44,868 samples; 27,371 are human-written and 17,497 are AI-generated. I split the dataset into train, dev, and test sets based on the labels. The dev and test sets each have 1,250 human-written essays and 1,250 AI-generated essays, resulting in 2,500 samples per evaluation split. The rest of the samples were used for training.

3.2.2 Balanced Split

After initially inspecting the DAIGT V2 Train Dataset, I found that the human-written and AI-generated classes were not equally distributed. The human-written side was dominated by the `persuade_corpus`, while the `train_essays` source contributed a smaller number of additional human-written essays. To reduce the effect of class imbalance, I created a balanced version of the dataset.

Before constructing the split, I removed three examples from `train_essays` that were labeled as AI-generated, since this source was otherwise treated as human-written essay data. I then separated the dataset into human-written and AI-generated pools. The AI-generated pool was kept in full, and the human-written pool was downsampled to match the total number of AI-generated examples. The balanced human pool includes all available `train_essays` examples and a random sample from `persuade_corpus`.

The dev and test sets were each constructed with 1,250 human-written and 1,250 AI-generated essays. AI-generated examples were sampled with a round-robin procedure across sources, while human-written examples were sampled from the balanced human pool. The remaining balanced examples were used for training. This split was designed to test model performance after controlling for class imbalance.

3.2.3 Source-Holdout Split

I created the source-holdout split to test whether models could generalize to unseen writing sources rather than simply learning source-specific artifacts. In this setting, some sources were excluded entirely from training and reserved only for evaluation.

For the human-written data, training and development examples were drawn from `persuade_corpus`, while held-out human evaluation examples were taken from `train_essays`. For the AI-generated data, both Claude sources, `darragh_claude_v6` and `darragh_claude_v7`, were held out for evaluation only. All other AI sources were used for training and development.

The dev set was constructed from the training data pools and contained 1,250 human-written and 1,250 AI-generated examples. The test set was constructed only from held-out sources and was balanced by taking the maximum equal number of human-written and AI-generated examples available from the held-out pools. AI-generated test examples were sampled using a round-robin procedure across the held-out Claude sources, while human-written test examples were sampled from `train_essays`. The remaining training examples were balanced by class and used for training.

This split was intended to evaluate whether models learned broader signals of AI-generated writing or relied primarily on source-specific patterns, however after conducting initial experiments, there was not much of a different in results between this split and the balanced split. I speculated that since the DAIGT V2 Train Dataset is a publicly available dataset, the models may have seen the data before and may have even been trained on it.

3.3 New Evaluation Dataset

I created a new dataset specifically for this project in order to evaluate model performance on unseen data that was meant to mimic high school through college level academic essays. The dataset consists of 24 original essays, including 12 human-written essays and 12 AI-generated essays. I created the AI-generated essays using GPT-5.4, Claude Sonnet 4.6, and Gemini 3 Flash.

The essays were written in response to six original argumentative prompts covering topics such as digital vs. print reading, planned vs. spontaneous travel, and morning vs. night productivity preferences. I split the positions of each argument so I better control the content to reduce bias. Essays were written to a target length of 300-400 words.

4 Human Annotation Study

To explore how well humans performed at this classification task, I conducted a human annotation study and compared human judgments with gold labels and evaluated inter-annotator agreement. I created an annotation scheme for the annotators as a guide for how to annotate the DAIGT V2 data. Two annotators independently labeled 200 essays sampled from the balanced DAIGT V2 test split as either human-written or AI-generated. The annotation analysis used normalized-text matching to align annotations with gold labels.

Inter-annotator agreement reached 90% overall agreement with a Cohen's κ score of approximately 0.80, indicating strong agreement between annotators. Individual annotator accuracy against gold labels ranged from 92% to 94%, while consensus annotations achieved approximately 97.8% accuracy.

The annotation experiments provide additional insight into the difficulty of distinguishing human-written and AI-generated essays and offered a useful comparison point for automated detection systems.

Metric	Value
Percent Agreement	90.0%
Cohen's κ	0.80
Annotator 1 Accuracy	92.0%
Annotator 2 Accuracy	94.0%
Consensus Accuracy	97.8%

Table 1: Human annotation agreement and accuracy results.

5 Methodology

5.1 Encoder Classifier

The first model I evaluated was the transformer-based encoder classifier RoBERTa-base. The model were implemented using the Hugging Face Transformers library and fine-tuned as binary sequence classifiers for distinguishing between human-written and AI-generated essays.

Each essay was tokenized using the corresponding pretrained tokenizer associated with the selected encoder model. Essays were truncated to a maximum sequence length of 512 tokens and padded to a fixed length during batching. Binary labels were assigned using the convention 0 = human-written and 1 = AI-generated.

Training was performed using the AdamW optimizer with a default learning rate of 2×10^{-5}, batch size of 8, and 3 training epochs. Dev set accuracy was evaluated after each epoch, and the best checkpoint was saved for final evaluation. Random seeds were fixed across experiments to improve reproducibility.

During evaluation, models produced logits over the two output labels, and predictions were generated using the maximum logit value. Evaluation scripts additionally computed confusion matrices, per-class recall, majority-class baselines, and source-specific accuracy statistics.

5.2 Zero-Shot Large Language Model Evaluation

I also evaluated several large language models in a zero-shot classification setting. The evaluated models included GPT-4.1 mini, GPT-5.4 mini, Claude Sonnet 4.6, Gemini 2.5 Flash, and Llama 3.1 8B Instruct.

For zero-shot experiments, the models received a classification prompt instructing them to determine whether a given essay was human-written or AI-generated. The prompt defined the two labels and instructed the model to return exactly one output label: `Human` or `AI`.

All zero-shot experiments were conducted using deterministic decoding settings with temperature set to 0.0. Additional parsing and normalization procedures were implemented to handle formatting inconsistencies across model responses. Responses that could not be normalized into valid labels were treated as unresolved predictions.

Separate inference wrappers were implemented for OpenAI, Anthropic, Gemini, Vertex AI, and local Llama evaluation pipelines. These wrappers standardized prompting behavior and evaluation procedures across providers while supporting retry logic and response normalization.

5.3 Fine-Tuning Experiments

I evaluated supervised fine-tuning for both closed-source and open-source language models. Fine-tuning experiments were conducted using GPT-4.1 mini, Gemini 2.5 Flash, and Llama 3.1 8B Instruct. There were no Claude models available for fine-tuning at the time of this project.

For OpenAI and Gemini fine-tuning experiments, training examples were converted into JSONL format containing system prompts, user essay content, and target assistant labels. Fine-tuning datasets were constructed from manually cleaned subsets of the balanced DAIGT V2 data. Multiple training subset sizes were explored during development, including smaller curated datasets and larger-scale training configurations.

For the training data used in fine-tuning these models, I implemented data cleaning to remove essays with suspicious template artifacts, placeholder metadata, truncated examples, duplicate essays, and essays outside a word count range. I also identified and removed essays with prompt leakage and potentially low-quality generated content. I then created subsets from this cleaned data pool for training. The dev and eval data used in these experiments was the same as the data used in the earlier classifier and zero-shot model runs for consistency.

Open-source fine-tuning experiments used Llama 3.1 8B Instruct with parameter-efficient fine-tuning (PEFT) and LoRA adapters. Training was performed using the TRL `SFTTrainer` framework with 4-bit quantization to reduce memory usage during training. LoRA adapters were applied to the transformer projection layers, including query, key, value, and output projection modules.

The Llama fine-tuning setup used prompts that asked the model to classify essays as either human-written or AI-generated and to return exactly one label. During evaluation, the fine-tuned adapters were loaded on top of the base Llama model for inference.

5.4 Evaluation Metrics

Model performance was evaluated primarily using classification accuracy. Additional evaluation metrics included precision, recall, F1-score, confusion matrices, and per-class recall statistics. Because several experiments involved balanced binary datasets, majority-class baseline accuracy was also reported for comparison.

Evaluation scripts additionally computed source-level accuracy statistics when source metadata was available. This analysis made it possible to examine how detector performance varied across different human-writing sources and AI-generation models.

For human annotation experiments, inter-annotator agreement was measured using percent agreement and Cohen's κ. Confidence-level agreement was additionally analyzed to examine annotator uncertainty across difficult classification cases.

5.5 Sanity Checks and Validation

I implemented several sanity checks throughout this project to validate experimental reliability and reduce the likelihood of inflated benchmark performance.

Exact-text overlap checks were performed between train, development, and test splits to identify duplicate essays shared across partitions.

Shuffled-label experiments were conducted in which training labels were randomly permuted prior to model training. These experiments produced near-chance performance, confirming that the classifiers were not exploiting unintended leakage artifacts within the implementation pipeline.

Additional validation experiments examined source distributions, class balance, and training-size effects. Source-holdout evaluation settings were specifically designed to test whether models generalized beyond the AI-generation sources observed during training. Together, these validation procedures strengthened confidence that reported performance reflected meaningful learning rather than trivial dataset artifacts.

6 Results

6.1 Benchmark Performance on DAIGT V2

Initial experiments on the balanced DAIGT V2 data produced extremely high classification performance across several models. Fine-tuned encoder classifiers achieved near-perfect accuracy, with RoBERTa-base producing the strongest overall benchmark results. Fine-tuned large language models also achieved very high in-domain performance, often exceeding 98% accuracy on the balanced test split.

These results initially suggested that distinguishing between human-written and AI-generated essays might be a relatively straightforward classification task. However, the consistently high benchmark performance across multiple architectures also raised concerns regarding potential dataset artifacts and benchmark-specific patterns that models could exploit.

Table 2 summarizes model performance across both the DAIGT V2 benchmark and the new out-of-domain evaluation dataset.

Model	DAIGT V2	New Dataset
GPT-5.4 mini ZS	88.84%	58.33%
GPT-4.1 mini ZS	71.88%	66.67%
Claude Sonnet 4.6 ZS	94.72%	**100.00%**
Gemini 2.5 Flash ZS	89.68%	87.50%
Llama 3.1 8B Instruct ZS	49.68%	50.00%
RoBERTa FT	**99.72%**	83.33%
GPT-4.1 mini FT	98.60%	70.83%
Gemini 2.5 Flash FT	99.32%	66.67%
Llama 3.1 8B Instruct FT	99.56%	66.67%

Table 2: Accuracy of all evaluated models on DAIGT V2 and the new student essay dataset. The new dataset consists of 24 essays: 12 human-written and 12 AI-generated.

Table 3 compares zero-shot and fine-tuned performance on the new dataset. In contrast to DAIGT V2, fine-tuning does not consistently improve performance and in some cases leads to lower accuracy.

Model	Zero-Shot	Fine-Tuned
GPT-4.1 mini	66.67%	70.83%
Gemini 2.5 Flash	**87.50%**	66.67%
Llama 3.1 8B	50.00%	66.67%

Table 3: Zero-shot vs. fine-tuned performance on the new dataset.

The benchmark results demonstrate that multiple models were capable of achieving extremely strong performance when trained on data from a publicly available dataset. RoBERTa-base achieved 99.7% accuracy on DAIGT V2, while fine-tuned GPT-4.1 mini, Gemini 2.5 Flash, and Llama 3.1 8B Instruct all exceeded 98% accuracy under fine-tuned conditions. Zero-shot closed-source models also performed strongly on the benchmark dataset, particularly Claude Sonnet 4.6 and Gemini 2.5 Flash.

Despite these strong benchmark results, performance on the new evaluation dataset was substantially more variable. Several models that achieved near-perfect benchmark accuracy experienced large performance drops when evaluated on the new dataset. This gap suggests that high benchmark accuracy alone may not reliably reflect how well these models perform on unseen data or in more realistic writing settings.

6.2 New Dataset Evaluation

Evaluation on the new essay dataset revealed significantly greater variation in model performance. While several models generalized reasonably well, others experienced significant reductions in accuracy compared to their benchmark performance.

Among zero-shot models, Claude Sonnet 4.6 achieved the strongest results, correctly classifying all essays in the evaluation set. Gemini 2.5 Flash also generalized relatively well, achieving 87.5% accuracy. In contrast, GPT-5.4 mini exhibited a strong bias toward predicting the AI-generated label, producing low recall on human-written essays despite strong benchmark performance.

Llama 3.1 8B Instruct performed poorly in the zero-shot setting and frequently defaulted toward predicting essays as human-written. The model achieved performance close to chance on the new dataset, highlighting substantial differences in zero-shot classification behavior across model families.

These results suggest that benchmark success does not necessarily correspond to strong generalization. Some models appeared highly sensitive to changes in prompt style, topic distribution, or generator source, while others generalized more effectively beyond the benchmark setting.

6.3 Paragraph-Level Experiments

Additional exploratory experiments evaluated Claude Sonnet 4.6 classification on paragraphs rather than full essay classification. A small custom dataset was constructed consisting of 20 unseen paragraphs, including 10 human-written paragraphs and 10 AI-generated paragraphs produced by Claude Sonnet 4.6.

The experiments compared classification performance using both the Claude API and the Claude graphical user interface (GUI). Table 4 summarizes the results.

Evaluation Setting	Human Accuracy	AI Accuracy
Claude API	100.00%	70.00%
Claude GUI	90.00%	80.00%

Table 4: Claude Sonnet 4.6 paragraph-level classification accuracy on unseen human-written and AI-generated paragraphs.

Performance remained relatively strong on human-written paragraphs but declined on AI-generated paragraphs compared to the full-essay experiments. These results suggest that shorter isolated passages may provide fewer stylistic and structural cues for reliable AI-generated text detection. The differences between API and GUI performance may additionally reflect variability in prompting behavior or interface-level response formatting.

6.4 Fine-Tuning Experiments

Fine-tuning consistently improved DAIGT V2 benchmark performance across nearly all evaluated models. Fine-tuned GPT-4.1 mini, Gemini 2.5 Flash, and Llama 3.1 8B Instruct all achieved substantial gains relative to their zero-shot counterparts on DAIGT V2.

However, improvements in benchmark accuracy did not consistently translate into comparable gains on the new dataset. Although fine-tuning generally improved overall performance, several fine-tuned models still exhibited reductions in performance under more realistic evaluation conditions.

For example, fine-tuned GPT-4.1 mini achieved approximately 98.6% benchmark accuracy but only 70.8% accuracy on the new evaluation dataset. Similarly, fine-tuned Gemini 2.5 Flash achieved approximately 99.3% benchmark accuracy but experienced lower performance on the new dataset than its benchmark results might predict. Gemini performance also dropped from 87.5% with zero-shot to 66.7% with fine-tuning, suggesting that fine-tuning may actually restrict models' ability to learn and adapt to new data. In contrast, RoBERTa-base maintained relatively strong performance across both settings, suggesting that encoder-based classifiers may remain competitive despite concerns regarding benchmark artifacts.

These results indicate that fine-tuning may partially improve performance while simultaneously increasing the risk of overfitting to benchmark-specific characteristics present in the training data.

6.5 Source-Holdout Evaluation

Source-holdout experiments provided additional evidence regarding model generalization behavior. In this setting, models were evaluated on AI-generation sources that were excluded entirely from training.

Performance remained relatively strong under source-holdout evaluation, although some degradation was observed compared to standard benchmark evaluation. The source-holdout experiments suggest that models learned at least some transferable characteristics associated with AI-generated writing rather than relying exclusively on memorization of model-specific artifacts.

However, the persistence of performance degradation under held-out conditions also indicates that detector behavior remains partially dependent on source-specific stylistic cues. These findings further support the importance of evaluating AI detectors under more challenging and distributionally distinct conditions.

Evaluation Setting	Accuracy
Balanced Split	99.72%
Source-Holdout Split	98.62%

Table 5: RoBERTa-base performance on balanced and source-holdout evaluation settings.

The human annotation experiments additionally showed that distinguishing between human-written and AI-generated essays remained challenging even for human evaluators, despite substantial inter-annotator agreement.

7 Discussion

The experiments demonstrate that high benchmark performance does not necessarily imply strong generalization for AI-generated essay detection. Although several models achieved near-perfect accuracy on the balanced DAIGT V2 benchmark, performance on the new dataset containing unseen data was substantially more variable. These findings suggest that benchmark datasets alone may provide an incomplete picture of detector reliability in realistic educational settings.

One important observation is that different model families generalized very differently beyond the benchmark setting. Claude Sonnet 4.6 maintained extremely strong performance under zero-shot evaluation, while several other models experienced substantial reductions in accuracy despite strong benchmark results. Similarly, the fine-tuned encoder classifier and large language models frequently achieved benchmark accuracies exceeding 98-99%, yet these improvements did not always transfer to the new evaluation dataset. This pattern suggests that some models may partially rely on benchmark-specific artifacts, source distributions, or stylistic regularities present within the training data.

The source-holdout experiments provide additional evidence that AI detection remains partially dependent on generator-specific characteristics. Although models retained relatively strong performance on held-out sources, measurable degradation persisted compared to standard benchmark evaluation. This finding indicates that detectors likely learn a combination of transferable linguistic patterns and source-specific stylistic cues.

The experiments also highlight important differences between zero-shot and fine-tuned detection approaches. Fine-tuning consistently improved benchmark performance, but the improvements observed on evaluation on unseen data were often considerably smaller. In some cases, fine-tuning may have increased sensitivity to benchmark-specific patterns while providing limited improvements in broader generalization. These results suggest that benchmark optimization alone may not produce detectors that remain reliable across evolving language models and differing types of writing.

The human annotation experiments further demonstrate the inherent difficulty of AI-generated text detection. Although annotators achieved strong overall agreement and accuracy, disagreement and uncertainty remained common across more ambiguous examples. This finding is important because educational discussions surrounding AI detection tools often assume that human-written and AI-generated text can be reliably distinguished. The annotation results suggest that the distinction is not always obvious, even for human evaluators.

Additional exploratory paragraph-level experiments also suggested that AI detection may become less reliable when models are evaluated on shorter isolated passages rather than full essays.

The findings have several implications for educational use of AI detection systems. Extremely high benchmark accuracy should not be interpreted as evidence that detectors are universally reliable in real-world academic contexts. Model performance may vary substantially depending on the types of prompts, writing styles, and language models encountered during evaluation. False positives remain a significant concern, particularly in high-stakes settings where incorrect classifications could affect students academically or professionally.

But the experiments do not suggest that AI detection is impossible. Several models generalized reasonably well beyond the benchmark setting, indicating that at least some detectable differences between human-written and AI-generated writing remain present. However, the variability observed across models and evaluation conditions suggests that AI detection should be treated as a probabilistic and evolving task rather than a solved classification problem.

As language models continue to improve, distinctions between human-written and AI-generated text may become increasingly difficult to detect reliably. Future research should prioritize evaluation on unseen, real-world data, continual reassessment of benchmark datasets, and investigation into how AI detection systems behave under changing writing conditions and newer language model generations.

8 Limitations

Several limitations should be considered when interpreting the results of this project.

The new evaluation dataset was relatively small, consisting of 24 essays total. Although the dataset was designed specifically to reduce benchmark overlap and provide more realistic evaluation conditions, the limited sample size restricts the extent to which the results can be generalized across broader educational writing contexts.

The evaluation datasets focused primarily on English argumentative essays. Detector behavior may differ substantially across other genres, writing styles, educational levels, or languages. Future work could extend the experiments to multilingual datasets and additional forms of academic writing.

The language model ecosystem continues to evolve rapidly. AI-generated writing produced by newer models may differ significantly from the outputs included in this study. As a result, detector performance observed during these experiments may not fully reflect future AI-generation capabilities.

The experiments also focused primarily on fully human-written and fully AI-generated essays. In realistic educational settings, student writing may involve varying degrees of AI assistance, editing, paraphrasing, or collaborative revision. Such hybrid writing scenarios may be significantly more difficult to classify reliably and were not extensively examined in this project.

Although the annotation experiments provided useful human comparison data, the annotation study involved only two annotators and a relatively small evaluation subset. Annotation studies on a larger scale could provide additional insight into how reliably humans distinguish between human-written and AI-generated essays.

Despite these limitations, the project provides evidence that benchmark performance alone may not adequately capture the reliability of AI detection systems under more realistic evaluation conditions.

9 Conclusion

This project evaluated the reliability of AI-generated essay detection across benchmark and out-of-domain settings using both encoder-based classifiers and large language models under zero-shot and fine-tuned conditions. While several models achieved extremely high performance on the DAIGT V2 benchmark dataset, these results did not always generalize reliably to new, unseen data.

The experiments showed substantial variation across model families and evaluation settings, suggesting that benchmark accuracy alone may overestimate detector reliability in realistic educational contexts. Overall, the findings highlight the importance of evaluation on unseen, source-diverse data and suggest that AI-generated text detection should be treated as an evolving and probabilistic task rather than a fully solved problem.

References

2025. Unmasking ai-generated texts. *International Journal of Advanced Computer Science and Applications*, 16(3).

Anthropic. 2025. Claude sonnet 4 technical report. https://www.anthropic.com.

Logan Burleigh. 2023. Aide: Ai detection for essays dataset. https://www.kaggle.com/datasets/lburleigh/tla-lab-ai-detection-for-essays-aide-dataset.

Scott Crossley et al. 2023. Persuade 2.0 corpus. https://github.com/scrosseye/persuade_corpus_2.0.

Team Gemini and Google DeepMind. 2024. Gemini: A family of highly capable multimodal models.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Yinhan Liu, Myle Ott, Naman Goyal, et al. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*.

OpenAI. 2025. Gpt-4.1 technical report. https://openai.com.

OpenAI. 2026. Gpt-5.4 technical report. https://openai.com.

Luka Terčon and Kaja Dobrovoljc. 2025. Linguistic characteristics of ai-generated text: A survey. *arXiv preprint arXiv:2510.05136*.

TheDRCat. 2024. Daigt v2 train dataset. https://www.kaggle.com/datasets/thedrcat/daigt-v2-train-dataset.

Daniel Yan, Katherine Blackwell, et al. 2023. Detection of gpt-generated essays in educational writing assessment. *Praxis der Textanalyse und Medienbewertung*.